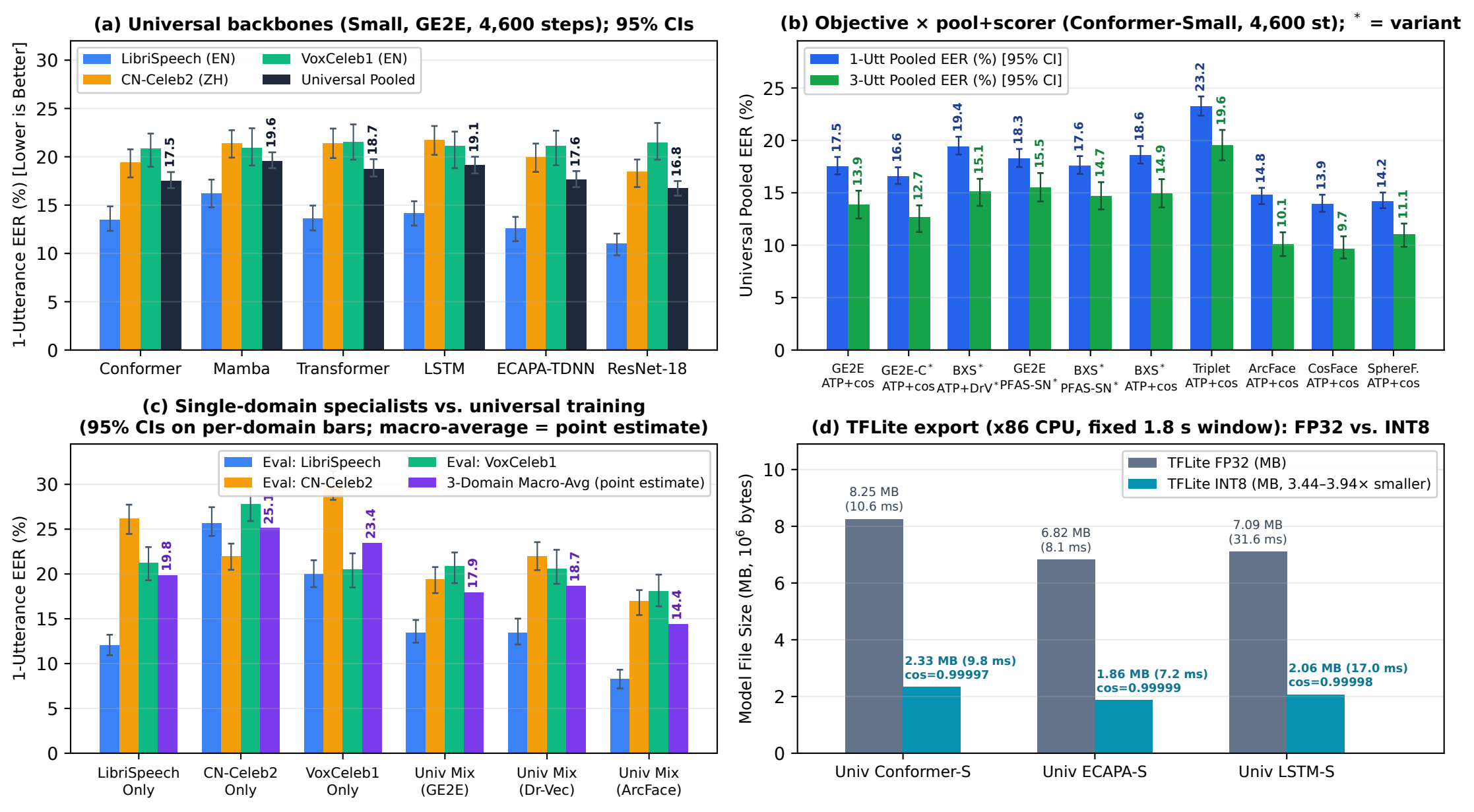
(a) Universal backbones (Small, GE2E, 4,600 steps); 95% CIs
1-Utterance EER (%) [Lower is Better]
LibriSpeech (EN)
CN-Celeb2 (ZH)
VoxCeleb1 (EN)
Universal Pooled
17.5
19.6
18.7
19.1
17.6
16.8
Conformer
Mamba
Transformer
LSTM
ECAPA-TDNN
ResNet-18
(b) Objective × pool+scorer (Conformer-Small, 4,600 st); * = variant
Universal Pooled EER (%)
1-Utt Pooled EER (%) [95% CI]
3-Utt Pooled EER (%) [95% CI]
17.5
13.9
16.6
12.7
19.4
15.1
18.3
15.5
17.6
14.7
18.6
14.9
23.2
19.6
14.8
10.1
13.9
9.7
14.2
11.1
GE2E ATP+cos
GE2E-C* ATP+cos
BXS* ATP+DrV*
GE2E PFAS-SN*
BXS* PFAS-SN*
BXS* ATP+cos
Triplet ATP+cos
ArcFace ATP+cos
CosFace ATP+cos
SphereF. ATP+cos
(c) Single-domain specialists vs. universal training
(95% CIs on per-domain bars; macro-average = point estimate)
1-Utterance EER (%)
Eval: LibriSpeech
Eval: CN-Celeb2
Eval: VoxCeleb1
3-Domain Macro-Avg (point estimate)
19.8
25.1
23.4
17.9
18.7
14.4
LibriSpeech Only
CN-Celeb2 Only
VoxCeleb1 Only
Univ Mix (GE2E)
Univ Mix (Dr-Vec)
Univ Mix (ArcFace)
(d) TFLite export (x86 CPU, fixed 1.8 s window): FP32 vs. INT8
Model File Size (MB, 10^6 bytes)
TFLite FP32 (MB)
TFLite INT8 (MB, 3.44–3.94× smaller)
8.25 MB (10.6 ms)
2.33 MB (9.8 ms) cos=0.99997
6.82 MB (8.1 ms)
1.86 MB (7.2 ms) cos=0.99999
7.09 MB (31.6 ms)
2.06 MB (17.0 ms) cos=0.99998
Univ Conformer-S
Univ ECAPA-S
Univ LSTM-S